

IAC ECOSYSTEM VIABILITY IMPLEMENTATION FRAMEWORK

Indicator Data Source, ZIP Availability, and Reliability Matrix

DATA FRAMEWORK NOTES

Purpose

This matrix identifies:

- recommended data sources,
- ZIP-level availability,
- reliability,
- and strategic usefulness

for each proposed IAC ecosystem indicator.

The framework is designed specifically for:

- rural/service ecosystem analysis,
- ecosystem transition monitoring,
- and distributed regional viability assessment.

Reliability Scale

Score Meaning

- | | |
|---|--|
| 5 | Highly reliable, standardized, strong geographic precision |
| 4 | Strong reliability with manageable limitations |
| 3 | Useful but partially modeled or interpolated |
| 2 | Weak granularity or inconsistent methodology |
| 1 | Experimental or limited comparability |

ZIP Availability Scale

Rating Meaning

- | | |
|----------|---|
| High | Direct ZIP/ZCTA-level availability |
| Moderate | Census tract/block-group interpolation required |
| Low | Mostly county/regional approximations |

SECTION I — NODE CONDITION INDICES

1. Human Continuity Index

Indicator	Primary Sources	ZIP Availability	Accuracy	Notes
Population Stability	ACS 5-Year Estimates	High	5	Core demographic foundation
Household Formation	ACS	High	5	Strong ZCTA availability
Youth Retention	IRS Migration Data, ACS	Moderate	4	Longitudinal estimation needed
Age Balance	ACS	High	5	Highly stable variable

Indicator	Primary Sources	ZIP Availability	Accuracy	Notes
Migration Dynamics	IRS SOI Migration Files	Moderate	4	Strong trend capability
Educational Continuity	ACS, State Education Agencies	Moderate	3	District overlay needed
Public Health Stability	CDC PLACES	High	4	Excellent emerging source
Social Continuity	ACS residency duration	Moderate	3	Proxy-based

2. Economic Function Index

Indicator	Primary Sources	ZIP Availability	Accuracy	Notes
Employment Stability	ACS, LEHD	Moderate	4	Strong labor data
Economic Diversity	County Business Patterns, NAICS	Moderate	3	Requires calculation
Local Business Vitality	Data Axle, CBP	Moderate	3	Commercial enhancement helpful
Income Sustainability	ACS	High	5	Strong comparability
Entrepreneurship	Nonemployer Statistics	Moderate	3	Lagged but useful
Economic Leakage	Consumer Expenditure + Retail Gravity Modeling	Low	2	Modeled indicator
Workforce Alignment	Lightcast/Burning Glass	Moderate	4	Commercial datasets strongest
Commercial Stability	USPS Vacancy Data, HUD	Moderate	3	Proxy-based

3. Infrastructure & Connectivity Index

Indicator	Primary Sources	ZIP Availability	Accuracy	Notes
Broadband Reliability	FCC Broadband Maps	High	4	Best national baseline
Transportation Connectivity	LEHD, DOT	Moderate	3	Travel-flow modeling useful
Housing Condition	ACS	High	4	Strong consistency
Utility Stability	State Utility Commissions	Low	2	Difficult nationally
Logistics Accessibility	Freight Analysis Framework, GIS	Moderate	3	Regional modeling required
Digital Participation	ACS Computer & Internet Use	High	4	Strong proxy
Infrastructure Age	Local assessor/state infrastructure records	Low	2	Fragmented availability

Indicator	Primary Sources	ZIP Availability	Accuracy	Notes
Environmental Infrastructure	EPA SDWIS, wastewater databases	Moderate	3	Utility-region based

4. Civic & Institutional Stability Index

Indicator	Primary Sources	ZIP Availability	Accuracy	Notes
Municipal Capacity	Census of Governments	Low	3	Municipality-based
Fiscal Stability	State audits, CAFRs	Low	4	Strong where available
Civic Participation	Census Voting Supplements	Low	2	Weak ZIP precision
Leadership Continuity	Local qualitative assessment	Low	2	Mostly qualitative
Institutional Density	IRS Nonprofit Registry	Moderate	4	Strong proxy
Volunteer Capacity	AmeriCorps/CNCS	Low	2	Sparse coverage
Intergovernmental Collaboration	Regional planning agencies	Low	2	Qualitative
Public Engagement/Cohesion	Survey + participation proxies	Low	2	Emerging methodology

SECTION II — SERVICE ECOSYSTEM INDICES

5. Essential Service Ecosystem Index

Indicator	Primary Sources	ZIP Availability	Accuracy	Notes
Healthcare Network Stability	HRSA, CMS, hospital databases	Moderate	4	Regional service modeling critical
Educational Accessibility	NCES, State Education Agencies	Moderate	4	GIS travel overlays useful
Grocery/Service Coverage	USDA Food Access Atlas	Moderate	4	Strong rural utility
Childcare Ecosystem	HHS Childcare Data	Moderate	3	Incomplete nationally
Emergency Service Redundancy	FEMA/NFIRS/local GIS	Moderate	3	Response modeling needed
Senior-Service Coverage	CMS Aging Services	Moderate	3	Regional overlay needed
Retail Sufficiency	Data Axle, ESRI Business Analyst	Moderate	3	Stronger commercially
Essential Service Accessibility	GIS travel-time analysis	Moderate	4	Powerful ecosystem indicator

6. Ecosystem Connectivity & Flow Index

Indicator	Primary Sources	ZIP Availability	Accuracy	Notes
Commuting Flows	LEHD LODES	High	5	Exceptional dataset
Service Gravity	Custom gravity modeling	Moderate	3	Core ecosystem metric
Workforce Circulation	LEHD	High	5	Strong movement patterns
Institutional Partnerships	Regional agency mapping	Low	2	Qualitative supplementation
Economic Inflow/Outflow	BEA regional flow analysis	Low	2	Requires modeling
Transportation Linkage	DOT/FHWA traffic flows	Moderate	4	Strong regional indicators
Service Dependency Patterns	Healthcare/school/service catchments	Moderate	3	Emerging analytical layer
Regional Connectivity	GIS network analysis	Moderate	4	High strategic value

7. Ecosystem Role & Embeddedness Index

Indicator	Primary Sources	ZIP Availability	Accuracy	Notes
Service Role Classification	IAC internal typology	High	4	Core framework asset
Regional Dependency Significance	LEHD + service modeling	Moderate	3	Network analysis required
Ecosystem Redundancy	Infrastructure/service mapping	Moderate	3	Important resilience metric
Node Criticality	Network centrality analysis	Moderate	4	High ecosystem value
Institutional Embeddedness	Nonprofit + agency relationships	Low	2	Qualitative overlay
Economic Embeddedness	Supply-chain/business linkage modeling	Low	2	Emerging capability
Regional Support Relationships	Regional governance mapping	Low	2	Human-defined initially
Specialized Functional Role	IAC classification system	High	4	Strong ecosystem variable

SECTION III — TRANSITION DYNAMICS INDICES

8. Transition Pressure Index

Indicator	Primary Sources	ZIP Availability	Accuracy	Notes
Aging Acceleration	ACS longitudinal analysis	High	5	Excellent trend variable
Institutional Exhaustion	Leadership/service turnover	Low	2	Qualitative
Healthcare Fragility	HRSA + hospital closure data	Moderate	4	Major rural signal
Retail Collapse Risk	USPS vacancy + retail trends	Moderate	3	Strong emerging indicator
Climate Exposure	FEMA, NOAA	High	5	Excellent spatial precision
Workforce Depletion	ACS, LEHD	Moderate	4	Strong trend analysis
Economic Transition Exposure	Industry concentration analysis	Moderate	4	Useful for vulnerability
Housing Instability Risk	ACS, HUD CHAS	High	5	Strong consistency

9. Adaptive Capacity Index

Indicator	Primary Sources	ZIP Availability	Accuracy	Notes
Broadband-Enabled Adaptation	FCC + ACS	High	4	Critical future variable
Regional Collaboration	Regional partnerships/planning orgs	Low	2	Mostly qualitative
Workforce Retraining	DOL/state workforce agencies	Moderate	3	Service-area approximations
Leadership Adaptability	Qualitative assessment	Low	2	Human-scored initially
Innovation Adoption	Business/technology surveys	Moderate	3	Commercial data useful
Institutional Flexibility	Governance/service integration analysis	Low	2	Emerging indicator
Digital Transition Readiness	ACS + broadband usage	High	4	Increasingly measurable
Service Integration Capacity	Shared-service agreements	Low	2	Human-curated initially

10. Strategic Opportunity Index

Indicator	Primary Sources	ZIP Availability	Accuracy	Notes
Regional Hub Potential	LEHD + service gravity analysis	Moderate	4	Strong strategic value
Shared-Service Opportunity	Institutional mapping	Low	2	Human-assisted analysis
Remote-Work Opportunity	Broadband + occupational structure	High	4	Excellent emerging metric
Infrastructure Modernization Leverage	Infrastructure age + grant mapping	Moderate	3	Good intervention targeting
Ecosystem Consolidation Potential	Service overlap/network analysis	Moderate	3	Advanced ecosystem layer
Partnership Readiness	Institutional participation mapping	Low	2	Qualitative supplementation
Economic Diversification Potential	Industry mix + workforce analysis	Moderate	4	Strong strategic metric
Transition Investment Readiness	Composite ecosystem analysis	Moderate	3	Higher-order synthesized metric

STRATEGIC OBSERVATIONS

1. The Framework Naturally Divides into Three Data Tiers

Tier Description

Tier 1 Direct ZIP/ZCTA observed data

Tier 2 GIS/interpolated/service-area modeled data

Tier 3 Human-curated ecosystem intelligence

All three are necessary.

2. Human-Defined Ecosystem Intelligence Remains Essential

Many of the most important ecosystem indicators cannot yet be fully automated.

- institutional exhaustion,
- leadership continuity,
- ecosystem cooperation,
- service integration,
- regional trust structures.

These will likely require hybrid quantitative + expert-scored systems. That is normal for advanced ecosystem intelligence frameworks.

3. The Strongest National Datasets

Dataset	Strategic Importance
ACS 5-Year Estimates	Demographic backbone
LEHD LODES	Workforce/network flows
CDC PLACES	Local health/social conditions
FCC Broadband Maps	Digital transition
HRSA/CMS	Healthcare ecosystem viability
FEMA/NOAA	Risk exposure
USDA Food Access Atlas	Rural service viability
IRS Migration Files	Population movement
HUD CHAS	Housing stress
EPA EJScreen	Environmental burden

FINAL RECOMMENDATION

The IAC framework should intentionally embrace mixed-method ecosystem intelligence. The future strength of the framework will come not from perfectly precise ZIP-level statistics, but from **integrating measurable conditions with functional ecosystem understanding**. That is the correct architecture for rural transition analysis, distributed service ecosystem planning, and long-duration ecosystem transition management.